

# **L2b:** Yield, Duration, Convexity, and the Yield Curve

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CHEME 5660 Cornell University

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# Scenario

It is 6:30 AM at your trading desk. You are responsible for a \$500 million portfolio of Treasury securities. The Federal Reserve meets today, and the rumor is a half-point move in the policy rate. The phone is already ringing.

By 2:00 PM you need an answer to one question: how much does this portfolio move?

The answer is conditional: it depends on the portfolio's cash flows and on which yields move. Today we estimate it from the slope and curvature of the price-yield relationship.

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**Careful:** a half-point move in the *policy rate* is not a half-point move in *your security's yield*. We return to that.

# Objectives for Today

We take the fixed-income price model from L2a and ask how its price responds when the discount rate changes.

## Today's concepts

- **Price versus yield**: explain why the price of a fixed positive cash-flow stream decreases with yield and follows a convex curve.
- **Duration and convexity**: define both measures, estimate the local price effect of a yield change, and interpret their units.
- **Yield-curve vocabulary**: distinguish yield to maturity, par yield, spot rates, and the discount factors that price individual cash flows.

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Companion notebook: [L2b lecture notebook](#)

# Lectures and Examples

**Lecture:** CHEME-5660-L2b-Lecture-Yield-Duration-Convexity-Curve-Fall-2026.ipynb

The required example prices a coupon Treasury security and studies its response to changes in the yield and in the coupon rate.

**Example:** CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb

The theorem exercises and term-structure notebooks remain optional extensions under advanced/.

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**Link:** <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

Quick review before we start: a safe payment can still have a risky price.

# Concept Review: Are Treasury Securities Really Risk-Free?

**No.** Treasuries carry low default risk as obligations of the U.S. government, which is why declared rates are so often taken from them. The label is conditional; two remaining risks matter today.

- **Default risk:** the issuer fails to pay. The United States has never repudiated its debt (a 1979 operational failure did delay some bill payments); other sovereigns have: Greece restructured in 2012, Russia defaulted in 1998.
- **Interest-rate risk:** a change in required yields changes the secondary-market value of fixed promised cash flows, hitting an institution that marks to market even without a sale.

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**Today:** interest-rate risk is the one we can compute. Its size depends on maturity, coupon, yield, and the shock.

# Case Study: Interest Rate Risk and Silicon Valley Bank

SVB held a large portfolio of **long-maturity fixed-rate** government securities bought while rates were low. As the Federal Reserve raised rates through 2022 and 2023, the value of those securities fell.

- The losses were unrealized, so a hold-to-maturity story obscured them.
- In early March 2023 uninsured depositors withdrew funds. SVB announced a capital raise to cover realized bond losses, and the bank failed.
- The FDIC took control on March 10, 2023, with roughly \$209 billion in assets: then the second largest failure of an insured bank on record, now the third.

Falling bond values and deposit withdrawals reinforced each other; interest-rate risk mattered, but was not the only cause.

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**Overview:** [the collapse of Silicon Valley Bank, from PBS](#)



# Malkiel's Bond-Pricing Theorems

Our destination: five classical results, stated now and proved as the tools arrive. Each assumes an **option-free** security with fixed positive cash flows, holds the other contract terms fixed, and compares maturities between bonds priced **at par** ( $c = y$ ).

1. Bond prices move **inversely** to bond yields.
2. **Longer maturity** means more sensitivity to a yield change.
3. The sensitivity gain from added maturity **diminishes** as maturity lengthens.
4. Equal yield moves produce **asymmetric** price changes: a decrease raises the price by more than an equal increase lowers it.
5. **Higher coupon rates** mean less sensitivity, in percentage terms.

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**Reference:** Malkiel (1962), *Expectations, Bond Prices, and the Term Structure of Interest Rates*, QJE 76(2), 197–218.

# Zero-Coupon Treasury Bill

Start with the simplest security: a bill pays nothing until maturity, then returns its face value.

Let  $V_P > 0$  be the face value,  $T > 0$  the remaining maturity in years,  $n$  the compounding intervals per year, and  $y$  the nominal annual yield. With per-period growth factor  $1 + y/n$ , the L2a discount factor is  $\mathcal{D}_{T,0}^{-1}(y; n) = (1 + y/n)^{-nT}$  and the price is:

$$V_B = \underbrace{(1 + y/n)^{-nT}}_{\text{discount}} V_P.$$

**Standing assumption:**  $y > 0$  throughout, as in the Treasury data we use; we flag every result that depends on it.

# Deriving the Price Sensitivity

Three inputs could move the price: the yield  $y$ , the coupon rate  $c$ , and the maturity  $T$ . A bill's contract fixes  $c = 0$ , so with  $V_P$  and  $n$  held fixed only  $y$  and  $T$  remain in the total differential:

$$dV_B = \underbrace{-V_P (1 + y/n)^{-nT} \left( \frac{T}{1 + y/n} \right) dy}_{V_{B,y}} - \underbrace{V_P (1 + y/n)^{-nT} (n \log(1 + y/n)) dT}_{V_{B,T}}.$$

**Why no  $dc$  term:** in the coupon-bearing family of the next section,  $\partial V_B / \partial c = (V_P / n) \sum_j \mathcal{D}_{j,0}^{-1}(y; n) > 0$ , strictly positive even at  $c = 0$ . The derivative is not zero; the contract requires  $dc = 0$ .

# What Does a Change in Maturity Mean?

A bill promises one payment, so  $T$  enters only through the exponent  $nT$  and  $\partial V_B / \partial T$  is well defined. But  $dT$  can describe two different comparisons:

- **Changing the contractual maturity:** a 6-month versus a 12-month bill compares securities issued on different terms. Smooth for a bill, where only the exponent changes; not for a coupon security, where it adds or removes scheduled payments.
- **The passage of time:** holding one bill, calendar time advances and remaining maturity falls, so  $dT = -dt$ . Smooth for a bill over the whole holding period; for a coupon security, only between coupon dates.

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**Later:** both distinctions return when we price coupon-bearing notes and bonds.

# Yield and Maturity Sensitivities

Read  $V_{B,y}$  and  $V_{B,T}$  as **magnitudes**: the derivatives are their negatives, hence the minus signs above. For  $y > 0$  both are positive:

$$V_{B,y} = \underbrace{V_P(1 + y/n)^{-nT}}_{=V_B} \underbrace{\left(\frac{T}{1 + y/n}\right)}_{>0}, \quad V_{B,T} = \underbrace{V_P(1 + y/n)^{-nT}}_{=V_B} \underbrace{(n \log(1 + y/n))}_{=g_y},$$

where  $g_y := n \log(1 + y/n)$  is the continuously compounded equivalent of  $y$  from L2a. With  $1 + y/n > 0$ , the first magnitude is positive for any  $T > 0$ . The second needs  $y > 0$ :  $g_y$  carries the sign of  $y$ , so at  $y = 0$  maturity stops mattering, and for  $-n < y < 0$  a longer bill would be worth *more*.

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**Careful:** negative nominal yields have occurred in other sovereign markets. State the condition.

# How Yield and Maturity Change a Bill's Price

## Decreasing in the yield

A higher yield discounts the face value more heavily, so  $V_B \downarrow$  as  $y \uparrow$ . Buy a 52-week bill at auction; if the yield moves from 2% to 3%, the secondary-market price falls below what you paid.

## Decreasing in the time to maturity

At a fixed positive yield,  $V_B \downarrow$  as  $T \uparrow$ . But while you *hold* the bill, remaining maturity falls,  $dT < 0$ : six months in, at an unchanged yield, the price is *higher* than you paid, because fewer periods remain to discount. At  $y = 0$ , every discount factor is one and the price does not move.

# Optional: Bill Maturity Sensitivity

The required lecture moves on to coupon-bearing notes and bonds; the advanced bill notebook sits here because it extends the maturity discussion just completed.

**Question:** How does a zero-coupon bill's price sensitivity change as its remaining maturity changes?

**Notebook:** advanced/malkiel/, the Treasury bill maturity example.

# Coupon-Bearing Treasury Notes and Bonds

Now let the security pay coupons. Let  $c$  be the annual coupon rate, so each payment is  $C := (c/n)V_P$ , and let  $N := nT$  count the remaining payment dates, indexed by  $j = 1, \dots, N$ . From L2a:

$$V_B = \mathcal{D}_{N,0}^{-1}(y; n) V_P + C \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n), \quad \mathcal{D}_{j,0}^{-1}(y; n) = (1 + y/n)^{-j}.$$

Holding that schedule fixed, the yield and coupon sensitivities are:

$$\frac{\partial V_B}{\partial y} = -\frac{1}{n(1 + y/n)} \left[ NV_P \mathcal{D}_{N,0}^{-1}(y; n) + C \sum_{j=1}^N j \mathcal{D}_{j,0}^{-1}(y; n) \right] < 0,$$

$$\frac{\partial V_B}{\partial c} = \frac{V_P}{n} \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n) > 0.$$

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**Careful:**  $N$  is a discrete schedule index. Differentiating a finite sum whose upper limit is  $nT$  with respect to a smooth  $T$  is not a valid total differential.



# How Bond Features Change Its Price

- **Decreasing in the yield:** now *both* the coupons and the final principal are discounted more heavily, so  $V_B \downarrow$  as  $y \uparrow$ .
- **Increasing in the coupon rate, but not proportional to it:**  $\partial V_B / \partial c$  does not depend on  $c$ , and the discounted par payment survives even at  $c = 0$ . Doubling the coupon doubles the coupon stream, not the price.
- **Maturity does not set the price direction by itself:** a **discount bond** ( $c < y$ ) drifts up toward par as maturity shortens, a **premium bond** ( $c > y$ ) drifts down toward par, and a **par bond** ( $c = y$ ) sits at par on a coupon date.

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**Careful:** Malkiel's maturity result is about *sensitivity to a yield change*, not about the price level.

# Duration and Convexity

Derivatives give the direction of a price change; risk management also needs its size, in units comparable across positions.

We assume an **option-free** security: its promised cash flows do not change when yields change. Conventional noncallable Treasuries qualify; callable, inflation-indexed, and mortgage-backed securities do not.

Let  $X_j \geq 0$  be the dollar cash flow at period  $j = 1, \dots, N$ , with at least one  $X_j > 0$ . One price formula covers bills and coupon securities alike:

$$V_B(y) = \sum_{j=1}^N X_j \mathcal{D}_{j,0}^{-1}(y; n).$$

A note pays  $X_j = C$  for  $j < N$  and  $X_N = C + V_P$ ; a bill pays  $X_j = 0$  for  $j < N$  and  $X_N = V_P$ .

# Two Derivatives, One Shape

Differentiate the price twice with respect to the yield, holding the cash flows  $X_j$  and the schedule length  $N$  fixed:

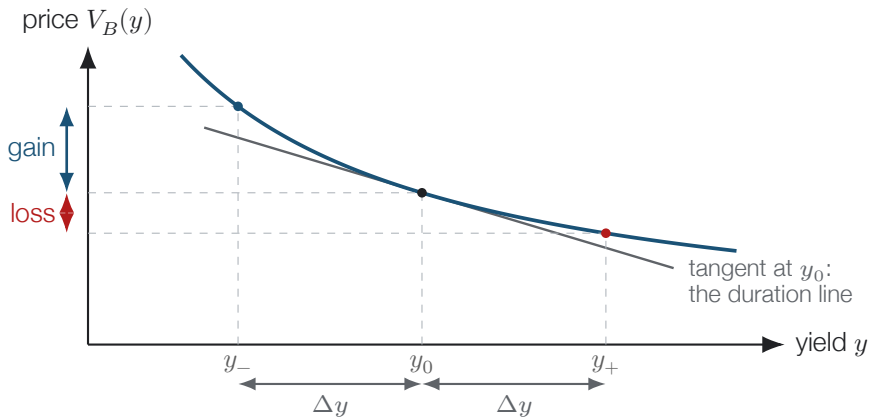
$$\frac{dV_B}{dy} = -\frac{1}{n} \sum_{j=1}^N j X_j (1 + y/n)^{-j-1} < 0,$$

$$\frac{d^2V_B}{dy^2} = \frac{1}{n^2} \sum_{j=1}^N j(j+1) X_j (1 + y/n)^{-j-2} > 0.$$

The strict signs follow from  $1 + y/n > 0$  and at least one  $X_j > 0$ : no term of either sum is negative, and at least one term of each is positive.

So the price-yield curve is **decreasing** and **strictly convex**. The second fact, **positive convexity**, is what makes the price response asymmetric.

# The Price-Yield Curve



Equal shocks  $y_{\pm} := y_0 \pm \Delta y$ : read on the price axis, the **gain** from the down-move exceeds the **loss** from the up-move. The curve lies above its tangent at  $y_0$ , so the duration line understates the gain and overstates the loss: convexity is the correction.

# The Two Risk Measures

Dividing each derivative by the price  $V_B(y) > 0$  removes position size and leaves two local risk measures, the **modified duration** and the **convexity**:

$$D_{\text{mod}}(y) := -\frac{1}{V_B(y)} \frac{dV_B}{dy}, \quad K(y) := \frac{1}{V_B(y)} \frac{d^2V_B}{dy^2}.$$

Both are positive by the signs just established. With  $y$  a decimal per year,  $D_{\text{mod}}$  has units of years and  $K$  units of years squared.

**Modified is not Macaulay.** The **Macaulay duration**  $D_{\text{mac}}$  is the present-value-weighted average time to payment, and  $D_{\text{mod}} = D_{\text{mac}}/(1 + y/n)$ . For a bill, every dollar arrives at maturity, so  $D_{\text{mac}} = T$  and  $D_{\text{mod}} = T/(1 + y/n)$ : exactly the ratio  $V_{B,y}/V_B$  derived earlier.

# Duration Gives the Slope; Convexity Adds Curvature

A second-order Taylor expansion about the current yield, divided by the price, turns the two measures into a repricing estimate. For a shock  $\Delta y$ , with

$$\Delta V_B := V_B(y + \Delta y) - V_B(y):$$

$$\frac{\Delta V_B}{V_B} \approx \underbrace{-D_{\text{mod}} \Delta y}_{\text{slope}} + \underbrace{\frac{1}{2}K(\Delta y)^2}_{\text{curvature}}.$$

Because  $K > 0$ , the curvature term is positive for a shock of *either* sign: it adds to the gain and subtracts from the loss.

**The convention.**  $\Delta y$  is a decimal change in the annual yield. One **basis point** is  $10^{-4}$ , so a 50-basis-point shock is  $\Delta y = 0.0050$ .

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**Careful:** a 50-basis-point move in the *policy rate* does not imply  $\Delta y = 0.0050$  for a given security.

# Estimate a Bond's Price Change

Take a note with  $D_{\text{mod}} = 7.2$  years and  $K = 65 \text{ years}^2$ , and shock its own yield by +50 basis points, so  $\Delta y = 0.0050$ :

- slope term:  $-7.2 \times 0.0050 = -0.0360$
- curvature term:  $\frac{1}{2} \times 65 \times (0.0050)^2 = +0.0008$
- estimate: a **-3.52%** price change, where the straight line alone would have said -3.60%

Both measures are **local**: derivatives at the current yield, so the estimate degrades as  $\Delta y$  grows. For moves the size of those that hurt SVB, reprice the whole schedule at the new yield rather than extrapolating from duration.

# What Duration Is Used For

Beyond the single-security estimate, duration is useful for two reasons:

- **A common scale:** securities with different coupons, maturities, and prices land on one scale, so the interest-rate risk of a 2-year note and a 10-year note can be compared directly.
- **A portfolio measure:** the duration of a portfolio is the market-value-weighted average of the durations of the securities it holds, so the \$500 million portfolio from the scenario reprices in one step rather than security by security.

The same properties support risk management: offsetting positions can bring a portfolio's net duration near zero, while a bank that funds long-duration securities with short-duration deposits, as SVB did, carries a duration mismatch.



# Revisiting Malkiel's Theorems

The machinery is in place: return to the five theorems from the start of the lecture. Ground rules: option-free, fixed positive cash flows, other terms held fixed, and maturity comparisons between bonds **selling at par**,  $c = y$ . The par restriction is not a technicality.

Substituting  $C = (c/n)V_P$  and summing the geometric series puts the coupon bond in closed form. Differentiating that at fixed  $c$ , then setting  $c = y$ , gives the modified duration of a par bond:

$$\frac{V_B}{V_P} = \frac{c}{y} + \left(1 - \frac{c}{y}\right) (1 + y/n)^{-N} \quad \Rightarrow \quad D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}.$$

**Order matters.** Differentiate at fixed  $c$ , *then* set  $c = y$ : the reverse differentiates along the par locus, where the price is pinned at  $V_P$ , and returns zero. Assume  $y > 0$  throughout.

# Malkiel's Theorems 1, 2, and 3

- **Theorem 1**, prices move inversely to yields: this is the sign  $dV_B/dy < 0$ , or equivalently  $D_{\text{mod}} > 0$ . Every promised payment is positive, so raising  $y$  discounts each more heavily.
- **Theorem 2**, longer maturity increases yield sensitivity:  
 $D_{\text{mod}}(N) = (1 - (1 + y/n)^{-N})/y$  is strictly increasing in  $N$ . A par bond starts at  $V_B = V_P$  at every maturity, so the dollar sensitivity  $-dV_B/dy = V_P D_{\text{mod}}(N)$  rises too: starting at par aligns the percentage and dollar rankings.
- **Theorem 3**, the gain diminishes: one extra period raises modified duration by  $D_{\text{mod}}(N + 1) - D_{\text{mod}}(N) = (1 + y/n)^{-N-1}/n$ , and successive increments shrink by the constant factor  $(1 + y/n)^{-1} < 1$ , so  $D_{\text{mod}}$  climbs toward the ceiling  $1/y$ .

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**Note:**  $1/y$  is the modified duration of a *perpetuity*, a security that pays its coupon forever and never repays principal.

# Malkiel's Theorems 4 and 5

- **Theorem 4**, price movements are asymmetric: a yield decrease raises the price by more than an equal increase lowers it, from the same starting yield. The reason is **positive convexity**. In the second-order estimate the slope term flips sign with the shock while the curvature term  $\frac{1}{2}K(\Delta y)^2$  stays positive. Nonlinearity alone would not do it: a concave curve gives the opposite asymmetry.
- **Theorem 5**, higher-coupon bonds are less sensitive: this is a statement about *percentage* sensitivity, and it needs  $N \geq 2$ . A larger coupon moves present value to earlier dates, lowering the weighted average payment time and so lowering  $D_{\text{mod}}$ . In *dollars* it runs the other way: more cash flow means a larger  $-dV_B/dy$ .

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**Excluded edge:** at  $N = 1$  everything arrives on one date and  $D_{\text{mod}} = 1/[n(1 + y/n)]$ , however large the coupon.

# Why Malkiel Compares Bonds at Par

Outside the par comparison, percentage and dollar sensitivities can rank securities differently. For a zero-coupon bill,  $D_{\text{mod}} = T/(1 + y/n)$  grows without bound, but the dollar sensitivity is a product of a growing and a shrinking factor:

$$-\frac{dV_B}{dy} = V_P \underbrace{\frac{T}{1 + y/n}}_{\text{growing}} \underbrace{(1 + y/n)^{-nT}}_{\text{shrinking}}.$$

So it rises, peaks, and falls. With  $V_P = 100$ ,  $n = 1$ ,  $y = 5\%$  it peaks near  $T \approx 20.5$  years at about 718 USD per unit of yield, then falls to about 541 by  $T = 40$ .

A 40-year bill is far more sensitive than a 20-year bill in **percentage** terms and *less* sensitive in **dollars**. Only the par comparison guarantees the two rankings agree.

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**Example:** CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb, which tests Theorems 4 and 5.

# From One Yield to the Yield Curve

**Yield to maturity** is the single internal rate that reproduces one security's price from its promised cash flows: useful for quoting and comparison, but not the full term structure.

A **spot rate** applies to one payment date, so a spot curve supplies one discount factor per date. A **par yield** is the coupon rate that makes a hypothetical security trade at par under that curve; the published Treasury yield curve is a **par yield curve**. Writing  $D(0, t_j)$  for the time-0 discount factor for maturity  $t_j$ , the curve-based price is:

$$V_B = \sum_{j=1}^N X_j D(0, t_j).$$

Level, slope, and curvature describe how discounting varies by maturity; by themselves they do not identify a forecast of future short rates.

# A Recession Predictor?

In most periods long-maturity yields exceed short-maturity yields, so the par yield curve slopes upward. When that ordering flips, the curve is **inverted**.

Inversions have a famous track record: an inverted long-minus-short spread, such as the **10-year minus 3-month** yield, has preceded every U.S. recession of the past half century.

The record comes with qualifiers: false positives occur, lead times run from months to years, and the reading depends on the maturity pair. The optional par-yield example computes the spread from Treasury data and asks what it says now.

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**Notebook:** [advanced/yield\\_curve/CHEME-5660-L3a-ParYieldCurves-Example-Fall-2026.ipynb](#)

# Optional Advanced Material

**Advanced 1:** [advanced/malkiel/](#), the Treasury bill maturity example and the par-bond duration note

**Advanced 2:** [advanced/yield\\_curve/](#), a lecture-style notebook with three examples

**Advanced 3:** [advanced/strips/](#), a lecture-style notebook and an issuer example

Optional, and not prerequisites for L3a. The malkiel pair takes Theorem 2 to auction data and proves Theorems 2 and 3 in closed form; the yield-curve sequence covers the investment yield of zeros, yield to maturity by the secant method, and par yield curves, including the recession-indicator spread; the STRIPS sequence infers spot, par, and implied forward rates from STRIPS prices.

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**Notebooks:** [lectures/week-2/L2b/advanced](#), also linked from the lecture notebook.

# Summary

This lecture developed the core tools for describing and managing the interest-rate risk of fixed-income securities.

## Key takeaways

- **Bond prices decrease as yields increase.** For a fixed stream of positive cash flows, the price-yield curve is decreasing and convex.
- **Duration and convexity measure local price sensitivity.** Duration is the first-order response to a yield change; convexity corrects the straight line for curvature.
- **Malkiel's theorems hold under stated assumptions.** The par restriction is what aligns the percentage and dollar sensitivity rankings.
- **One yield is not a yield curve.** Spot rates value each payment date, and par yields restate that curve as coupon rates.



That's a wrap. What are some of the interesting things we discussed today?